Differential Equations Workbook Solutions

Partial differential equations

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SEPARATION OF VARIABLES

■ 1. Solve the the partial differential equation by separating variables and assuming the separation constant is negative.

$$\frac{1}{k^2}\left(\frac{\partial^2 u}{\partial x^2}\right)+\frac{\partial u}{\partial t}=0$$

Solution:

We'll find the first and second derivatives of the product solution.

$$u(x,t)=v(x)w(t)$$

$$\frac{\partial u}{\partial t}=v(x)w'(t)$$

$$\frac{\partial u}{\partial x}=v'(x)w(t) \qquad \frac{\partial^2 u}{\partial x^2}=v''(x)w(t)$$

Then we'll plug the derivatives into the partial differential equation.

$$\frac{1}{k^2}v''(x)w(t)+v(x)w'(t)=0$$

$$\frac{1}{k^2}v''(x)w(t)=-v(x)w'(t)$$

$$\frac{1}{k^2}\left(\frac{d^2 v}{dx^2}\right)w(t)=-v(x)\frac{dw}{dt}$$

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$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{k^2}{w(t)}\right)\frac{dw}{dt}$$

The equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{k^2}{w(t)}\right)\frac{dw}{dt} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

$$r^2 + \lambda = 0$$

$$r = \pm\sqrt{-\lambda}$$

$$v(x) = c_1\cos(\sqrt{\lambda}x) + c_2\sin(\sqrt{\lambda}x)$$

and

$$-\left(\frac{k^2}{w(t)}\right)\frac{dw}{dt} = -\lambda$$

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$$\frac{dw}{dt} = \left(\frac{\lambda}{k^2}\right) w$$

$$\ln w = \left(\frac{\lambda}{k^2}\right) t + C$$

$$w = Ce^{\frac{\lambda t}{k^2}}$$

Then

$$u(x,t) = c_1 e^{\frac{\lambda t}{k^2}} \cos(\sqrt{\lambda} x) + c_2 e^{\frac{\lambda t}{k^2}} \sin(\sqrt{\lambda} x)$$

■ 2. Solve the the partial differential equation by separating variables and assuming the separation constant is positive.

$$16u_{yy} + u_{xx} = 0$$

Solution:

We'll find the first and second derivatives of the product solution.

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = v'(x)w(y) \qquad \frac{\partial u}{\partial y} = v(x)w'(y)$$

$$\frac{\partial^2 u}{\partial x^2} = v''(x)w(y) \qquad \frac{\partial^2 u}{\partial y^2} = v(x)w''(y)$$

Then we'll plug the derivatives into the partial differential equation.

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$$16v(x)w''(y) + v''(x)w(y) = 0$$

$$16v(x)w''(y) = -v''(x)w(y)$$

$$-\frac{16w''(y)}{w(y)} = \frac{v''(x)}{v(x)}$$

The equation is telling us that a function in terms of x will always be equivalent to a function in terms of y, regardless of the values of x and y that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

$$-\frac{16w''(y)}{w(y)} = \frac{v''(x)}{v(x)} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$-\left(\frac{16}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$-16\left(\frac{d^2w}{dy^2}\right) + \lambda w = 0$$

$$-16r^2 + \lambda = 0$$

$$r = \pm\frac{\sqrt{\lambda}}{4}$$

$$w(y) = c_1\cos\left(\frac{\sqrt{-\lambda}}{4}y\right) + c_2\sin\left(\frac{\sqrt{-\lambda}}{4}y\right)$$

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and

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

$$r^2 + \lambda = 0$$

$$r = \pm\sqrt{-\lambda}$$

$$v(x) = c_3 e^{\sqrt{-\lambda}x} + c_4 e^{-\sqrt{-\lambda}x}$$

Then

$$u(x,y) = \left(c_1 \cos\left(\frac{\sqrt{-\lambda}}{4}y\right) + c_2 \sin\left(\frac{\sqrt{-\lambda}}{4}y\right)\right)(c_3 e^{\sqrt{-\lambda}x} + c_4 e^{-\sqrt{-\lambda}x})$$

■ 3. Solve the the partial differential equation by separating variables and assuming the separation constant is negative.

$$9\frac{\partial^2 u}{\partial x^2} - 4\frac{\partial^2 u}{\partial y^2} = 0$$

Solution:

We'll find the first and second derivatives of the product solution.

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$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = v'(x)w(y) \qquad \frac{\partial u}{\partial y} = v(x)w'(y)$$

$$\frac{\partial^2 u}{\partial x^2} = v''(x)w(y) \qquad \frac{\partial^2 u}{\partial y^2} = v(x)w''(y)$$

Then we'll plug the derivatives into the partial differential equation.

$$9v''(x)w(y) - 4v(x)w''(y) = 0$$

$$9v''(x)w(y) = 4v(x)w''(y)$$

$$\frac{v''(x)}{v(x)} = \frac{4}{9}\left(\frac{w''(y)}{w(y)}\right)$$

The equation is telling us that a function in terms of x will always be equivalent to a function in terms of y, regardless of the values of x and y that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

$$\frac{v''(x)}{v(x)} = \frac{4}{9}\left(\frac{w''(y)}{w(y)}\right)$$

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{4}{9w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

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$$\frac{d^2v}{dx^2} + \lambda v = 0$$

$$r^2 + \lambda = 0$$

$$r = \pm\sqrt{-\lambda}$$

$$v(x) = c_1 \cos\left(\sqrt{\lambda}x\right) + c_2 \sin\left(\sqrt{\lambda}x\right)$$

and

$$\left(\frac{4}{9w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$4\frac{d^2w}{dy^2} + 9\lambda w = 0$$

$$4r^2 + 9\lambda = 0$$

$$r = \pm\frac{3\sqrt{-\lambda}}{2}$$

$$w(y) = c_3 \cos\left(\frac{3\sqrt{\lambda}}{2}y\right) + c_4 \sin\left(\frac{3\sqrt{\lambda}}{2}y\right)$$

Then

$$u(x,y) = \left(c_1 \cos\left(\sqrt{\lambda}x\right) + c_2 \sin\left(\sqrt{\lambda}x\right)\right)\left(c_3 \cos\left(\frac{3\sqrt{\lambda}}{2}y\right) + c_4 \sin\left(\frac{3\sqrt{\lambda}}{2}y\right)\right)$$

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■ 4. Solve the the partial differential equation by separating variables and assuming the separation constant is positive.

$$-\frac{\partial^2 u}{\partial x^2} + 25\frac{\partial^2 u}{\partial t^2} = 0$$

Solution:

We'll find the first and second derivatives of the product solution.

$$u(x,t) = v(x)w(t)$$

$$\frac{\partial u}{\partial x} = v'(x)w(t) \qquad \frac{\partial u}{\partial t} = v(x)w'(t)$$

$$\frac{\partial^2 u}{\partial x^2} = v''(x)w(t) \qquad \frac{\partial^2 u}{\partial t^2} = v(x)w''(t)$$

Then we'll plug the derivatives into the partial differential equation.

$$-v''(x)w(t) + 25v(x)w''(t) = 0$$

$$-v''(x)w(t) = -25v(x)w''(t)$$

$$\left(\frac{1}{v(x)}\right)\frac{d^2 v}{dx^2} = \left(\frac{25}{w(t)}\right)\frac{d^2 w}{dt^2}$$

The equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

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$$-\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{25}{w(t)}\right)\frac{d^2w}{dt^2} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

$$r^2 + \lambda = 0$$

$$r = \pm\sqrt{-\lambda}$$

$$v(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$$

and

$$\left(\frac{25}{w(t)}\right)\frac{d^2w}{dt^2} = -\lambda$$

$$25\frac{d^2w}{dt^2} + \lambda w = 0$$

$$25r^2 + \lambda = 0$$

$$r = \pm\frac{\sqrt{-\lambda}}{5}$$

$$w(t) = c_3 e^{\frac{\sqrt{-\lambda}}{5}t} + c_4 e^{-\frac{\sqrt{-\lambda}}{5}t}$$

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Then

$$u(x,y) = (c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x})(c_3 e^{\frac{\sqrt{-\lambda}}{5}t} + c_4 e^{-\frac{\sqrt{-\lambda}}{5}t})$$

■ 5. Solve the the partial differential equation by separating variables and assuming the separation constant is positive.

$$-\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} = 0$$

Solution:

We'll find the first and second derivatives of the product solution.

$$u(x,t) = v(x)w(t)$$

$$\frac{\partial u}{\partial t} = v(x)w'(t)$$

$$\frac{\partial u}{\partial x} = v'(x)w(t) \qquad \frac{\partial^2 u}{\partial x^2} = v''(x)w(t)$$

Then we'll plug the derivatives into the partial differential equation.

$$-v''(x)w(t) + v(x)w'(t) = 0$$

$$-v''(x)w(t) = -v(x)w'(t)$$

$$-\frac{d^2 v}{dx^2}w(t) = -v(x)\frac{dw}{dt}$$

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$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{1}{w(t)}\right)\frac{dw}{dt}$$

The equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{1}{w(t)}\right)\frac{dw}{dt} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \lambda$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

$$r^2 + \lambda = 0$$

$$r = \pm\sqrt{-\lambda}$$

$$v(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

and

$$\left(\frac{1}{w(t)}\right)\frac{dw}{dt} = -\lambda$$

$$\frac{dw}{dt} = -\lambda w$$

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$$\ln w = -\lambda t + C$$

$$w = Ce^{-\lambda t}$$

Then

$$u(x,t) = Ce^{-\lambda t}(c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x})$$

$$u(x,t) = c_1 e^{\sqrt{-\lambda}x-\lambda t} + c_2 e^{-\sqrt{-\lambda}x-\lambda t}$$

■ 6. Solve the the partial differential equation by separating variables and assuming the separation constant is negative.

$$\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial y^2} = 0$$

Solution:

We'll find the first and second derivatives of the product solution.

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = v'(x)w(y) \qquad \frac{\partial u}{\partial y} = v(x)w'(y)$$

$$\frac{\partial^2 u}{\partial x^2} = v''(x)w(y) \qquad \frac{\partial^2 u}{\partial y^2} = v(x)w''(y)$$

Then we'll plug the derivatives into the partial differential equation.

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$$v'(x)w(y) + v(x)w''(y) = 0$$

$$v'(x)w(y) = -v(x)w''(y)$$

$$\frac{dv}{dx}w(y) = -v(x)\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v(x)}\right)\frac{dv}{dx} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2}$$

The equation is telling us that a function in terms of x will always be equivalent to a function in terms of y, regardless of the values of x and y that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

$$\left(\frac{1}{v(x)}\right)\frac{dv}{dx} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{v(x)}\right)\frac{dv}{dx} = -\lambda$$

$$\frac{dv}{dx} = -\lambda v$$

$$\ln v = -\lambda x + C$$

$$v = Ce^{-\lambda x}$$

and

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$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

$$r^2 - \lambda = 0$$

$$r = \pm\sqrt{\lambda}$$

$$w(y) = c_1 e^{\sqrt{\lambda}y} + c_2 e^{-\sqrt{\lambda}y}$$

Then

$$u(x,y) = Ce^{\lambda x}(c_1 e^{\sqrt{\lambda}y} + c_2 e^{-\sqrt{\lambda}y})$$

$$u(x,y) = c_1 e^{\sqrt{\lambda}y - \lambda x} + c_2 e^{-\sqrt{\lambda}y - \lambda x}$$

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BOUNDARY VALUE PROBLEMS

■ 1. Solve the boundary value problem, if $y(0) = 3$ and $y'(0) = -4$.

$$y'' + 4y = 0$$

Solution:

Solving the associated characteristic equation gives

$$r^2 + 4 = 0$$

$$r^2 = -4$$

$$r = \pm 2i$$

which means the general solution to the homogeneous equation is

$$y(t) = c_1 \cos(2t) + c_2 \sin(2t)$$

Substituting $y(0) = 3$ into the general solution gives

$$3 = c_1 \cos(2 \cdot 0) + c_2 \sin(2 \cdot 0)$$

$$3 = c_1(1) + c_2(0)$$

$$c_1 = 3$$

Then we'll have

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$$y(t) = 3\cos(2t) + c_2 \sin(2t)$$

Now find $y'(t)$ and substitute $y'(0) = -4$.

$$y'(t) = -6\sin(2t) + 2c_2 \cos(2t)$$

$$-4 = -6\sin(2 \cdot 0) + 2c_2 \cos(2 \cdot 0)$$

$$-4 = 2c_2$$

$$c_2 = -2$$

Then the solution to the boundary value problem is

$$y(t) = 3\cos(2t) - 2\sin(2t)$$

■ 2. Solve the boundary value problem, if $y(0) = 1$, $y'(0) = 0$, and $y''(0) = 5$.

$$y''' - y'' + 9y' - 9y = 0$$

Solution:

Solving the associated characteristic equation gives

$$r^3 - r^2 + 9r - 9 = 0$$

$$r^2(r - 1) + 9(r - 1) = 0$$

$$(r - 1)(r^2 + 9) = 0$$

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$$r = 1, \pm 3i$$

which means the general solution to the homogeneous equation is

$$y(t) = c_1 e^t + c_2 \cos(3t) + c_3 \sin(3t)$$

$$y'(t) = c_1 e^t - 3c_2 \sin(3t) + 3c_3 \cos(3t)$$

$$y''(t) = c_1 e^t - 9c_2 \cos(3t) - 9c_3 \sin(3t)$$

Substituting $y(0) = 1$, $y'(0) = 0$, and $y''(0) = 5$ into the general solution gives

$$c_1 + c_2 = 1$$

$$c_1 + 3c_3 = 0$$

$$c_1 - 9c_2 = 5$$

Solving this system gives $c_1 = 7/5$, $c_2 = -2/5$, and $c_3 = -7/15$. Then the solution to the boundary value problem is

$$y(t) = \frac{7}{5}e^t - \frac{2}{5}\cos(3t) - \frac{7}{15}\sin(3t)$$

■ 3. Solve the boundary value problem, if $u(\pi/3,t) = 2e^{3t}$, $(\partial u/\partial x)(\pi/3,t) = -3e^{3t}$, and $-\lambda = -9$.

$$\frac{\partial^2 u}{\partial x^2} + 3\frac{\partial u}{\partial t} = 0$$

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Solution:

We'll find the first and second derivatives of the product solution.

$$u(x,t) = v(x)w(t)$$

$$\frac{\partial u}{\partial t} = v(x)w'(t)$$

$$\frac{\partial u}{\partial x} = v'(x)w(t) \qquad \frac{\partial^2 u}{\partial x^2} = v''(x)w(t)$$

Then we'll plug the derivatives into the partial differential equation.

$$v''(x)w(t) + 3v(x)w'(t) = 0$$

$$v''(x)w(t) = -3v(x)w'(t)$$

$$\frac{d^2v}{dx^2}w(t) = -3v(x)\frac{dw}{dt}$$

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{3}{w(t)}\right)\frac{dw}{dt}$$

The equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{3}{w(t)}\right)\frac{dw}{dt} = -9$$

Now that we have a three-part equation, we can break it apart into two separate equations.

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$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -9$$

$$\frac{d^2v}{dx^2} + 9v = 0$$

$$r^2 + 9 = 0$$

$$r = \pm 3i$$

$$v(x) = c_1 \cos(3x) + c_2 \sin 93x)$$

and

$$-\left(\frac{3}{w(t)}\right)\frac{dw}{dt} = -9$$

$$\left(\frac{1}{w(t)}\right)\frac{dw}{dt} = 3$$

$$\ln w = 3t + C$$

$$w = Ce^{3t}$$

Then

$$u(x,t) = c_1 e^{3t}\cos(3x) + c_2 e^{3t}\sin(3x)$$

$$\frac{\partial u}{\partial x}(x,t) = -3c_1 e^{3t}\sin(3x) + 3c_2 e^{3t}\cos(3x)$$

Substituting $u = (\pi/3,t) = 2e^{3t}$ and $(\partial u/\partial x)(\pi/3,t) = -3e^{3t}$ into the general solution gives

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$$-c_1 e^{3t} = 2e^{3t}$$

$$-3c_2 e^{3t} = -3e^{3t}$$

Then we'll have

$$c_1 = -2$$

$$c_2 = 1$$

Then the solution to the boundary value problem is

$$u(x,t) = -2e^{3t}\cos(3x) + e^{3t}\sin(3x)$$

■ 4. Solve the boundary value problem, if $u(\pi, t) = 0$, $u(x,0) = -2\cos(x/2)$, $(\partial u/\partial x)(\pi, t) = 2e^{-t} - e^{t}$, $(\partial u/\partial t)(x,0) = 6\cos(x/2)$, and $-\lambda = -\frac{1}{4}$.

$$4\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial t^2} = 0$$

Solution:

We'll find the first and second derivatives of the product solution.

$$u(x,t) = v(x)w(t)$$

$$\frac{\partial u}{\partial x} = v'(x)w(t) \qquad \frac{\partial^2 u}{\partial x^2} = v''(x)w(t)$$

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$$\frac{\partial u}{\partial t} = v(x)w'(t) \qquad \frac{\partial^2 u}{\partial t^2} = v(x)w''(t)$$

Then we'll plug the derivatives into the partial differential equation.

$$4v''(x)w(t) + v(x)w''(t) = 0$$

$$v''(x)w(t) = -\frac{1}{4}v(x)w''(t)$$

$$\frac{d^2v}{dx^2}w(t) = -\frac{1}{4}v(x)\frac{d^2w}{dt^2}$$

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{4w(t)}\right)\frac{d^2w}{dt^2}$$

The equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{4w(t)}\right)\frac{d^2w}{dt^2} = -\frac{1}{4}$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\frac{1}{4}$$

$$4\frac{d^2v}{dx^2} + v = 0$$

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$$4r^2 + 1 = 0$$

$$r = \pm \frac{1}{2}i$$

$$v(x) = c_1 \cos\left(\frac{x}{2}\right) + c_2 \sin\left(\frac{x}{2}\right)$$

and

$$-\left(\frac{1}{4w(t)}\right)\frac{d^2w}{dt^2} = -\frac{1}{4}$$

$$\frac{d^2w}{dt^2} - w = 0$$

$$r^2 - 1 = 0$$

$$r = \pm 1$$

$$w(t) = c_3 e^t + c_4 e^{-t}$$

Then

$$u(x,t) = \left(c_1 \cos\left(\frac{x}{2}\right) + c_2 \sin\left(\frac{x}{2}\right)\right)(c_3 e^t + c_4 e^{-t})$$

$$\frac{\partial u}{\partial x}(x,t) = \left(-\frac{1}{2}c_1 \sin\left(\frac{x}{2}\right) + \frac{1}{2}c_2 \cos\left(\frac{x}{2}\right)\right)(c_3 e^t + c_4 e^{-t})$$

$$\frac{\partial u}{\partial t}(x,t) = \left(c_1 \cos\left(\frac{x}{2}\right) + c_2 \sin\left(\frac{x}{2}\right)\right)(c_3 e^t - c_4 e^{-t})$$

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Substituting $u(\pi, t) = 0$, $u(x,0) = -2\cos(x/2)$, $(\partial u/\partial x)(\pi, t) = 2e^{-t} - e^{t}$, and $(\partial u/\partial t)(x,0) = 6\cos(x/2)$ into the general solution gives

$$c_2(c_3e^{t} + c_4e^{-t}) = 0$$

$$\left(c_1\cos\left(\frac{x}{2}\right) + c_2\sin\left(\frac{x}{2}\right)\right)(c_3 + c_4) = -2\cos\left(\frac{x}{2}\right)$$

$$-\frac{1}{2}c_1(c_3e^{t} + c_4e^{-t}) = 2e^{-t} - e^{t}$$

$$\left(c_1\cos\left(\frac{x}{2}\right) + c_2\sin\left(\frac{x}{2}\right)\right)(c_3e - c_4) = 6\cos\left(\frac{x}{2}\right)$$

If we choose $c_1 = 2$, we find $c_1 = 2$, $c_2 = 0$, $c_3 = 1$, and $c_4 = -2$. Then the solution to the boundary value problem is

$$u(x, t) = 2\cos\left(\frac{x}{2}\right)(e^{t} - 2e^{-t})$$

■ 5. Solve the boundary value problem, if $u(0,y) = e^{8y} + 3e^{-8y}$, $u(x,0) = 8e^{2x} - 4e^{-2x}$, $(\partial u/\partial x)(0,y) = 6e^{8y} + 18e^{-8y}$, $(\partial u/\partial y)(x,0) = 16e^{-2x} - 32e^{2x}$, and $-\lambda = 4$.

$$\frac{\partial^2 u}{\partial y^2} - 16\frac{\partial^2 u}{\partial x^2} = 0$$

Solution:

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We'll find the first and second derivatives of the product solution.

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = v'(x)w(y) \qquad \frac{\partial^2 u}{\partial x^2} = v''(x)w(y)$$

$$\frac{\partial u}{\partial y} = v(x)w'(y) \qquad \frac{\partial^2 u}{\partial y^2} = v(x)w''(y)$$

Then we'll plug the derivatives into the partial differential equation.

$$v(x)w''(y) - 16v''(x)w(y) = 0$$

$$-16v''(x)w(y) = -v(x)w''(y)$$

$$-16\frac{d^2v}{dx^2}w(y) = -v(x)\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{1}{16w(y)}\right)\frac{d^2w}{dy^2}$$

The equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda = 4$.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \left(\frac{1}{16w(y)}\right)\frac{d^2w}{dy^2} = 4$$

Now that we have a three-part equation, we can break it apart into two separate equations.

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$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = 4$$

$$\frac{d^2v}{dx^2} - 4v = 0$$

$$r^2 - 4 = 0$$

$$r = \pm 2$$

$$v(x) = c_1e^{2x} + c_2e^{-2x}$$

and

$$\left(\frac{1}{16w(y)}\right)\frac{d^2w}{dy^2} = 4$$

$$\frac{d^2w}{dy^2} - 64w = 0$$

$$r^2 - 64 = 0$$

$$r = \pm 8$$

$$w(y) = c_3e^{8y} + c_4e^{-8y}$$

Then

$$u(x, y) = (c_1e^{2x} + c_2e^{-2x})(c_3e^{8y} + c_4e^{-8y})$$

$$\frac{\partial u}{\partial x}(x, y) = (2c_1e^{2x} - 2c_2e^{-2x})(c_3e^{8y} + c_4e^{-8y})$$

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$$\frac{\partial u}{\partial y}(x,y) = (c_1 e^{2x} + c_2 e^{-2x})(8c_3 e^{8y} - 8c_4 e^{-8y})$$

Substituting $u(0,y) = e^{8y} + 3e^{-8y}$, $u(x,0) = 8e^{2x} - 4e^{-2x}$, $(\partial u/\partial x)(0,y) = 6e^{8y} + 18e^{-8y}$, and $(\partial u/\partial y)(x,0) = 16e^{-2x} - 32e^{2x}$ into the general solution gives

$$(c_1 + c_2)(c_3 e^{8y} + c_4 e^{-8y}) = e^{8y} + 3e^{-8y}$$

$$(c_1 e^{2x} + c_2 e^{-2x})(c_3 + c_4) = 8e^{2x} - 4e^{-2x}$$

$$(2c_1 - 2c_2)(c_3 e^{8y} + c_4 e^{-8y}) = 6e^{8y} + 18e^{-8y}$$

$$(c_1 e^{2x} + c_2 e^{-2x})(8c_3 - 8c_4) = 16e^{-2x} - 32e^{2x}$$

Then we'll have

$$(c_1 + c_2)c_3 = 1$$

$$(c_1 + c_2)c_4 = 3$$

$$2(c_1 - c_2)c_3 = 6$$

$$2(c_1 - c_2)c_4 = 18$$

If we solve this system, we find $c_1 = 2$, $c_2 = -1$, $c_3 = 1$, and $c_4 = 3$. Then the solution to the boundary value problem is

$$u(x,y) = (2e^{2x} - e^{-2x})(e^{8y} + 3e^{-8y})$$

■ 6. Solve the boundary value problem, if $u = (0,y) = 4\sin(\sqrt{3\lambda}y)$ and the separation constant is negative.

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$$3\frac{\partial u}{\partial x} - \frac{\partial^2 u}{\partial y^2} = 0$$

Solution:

We'll find the first and second derivatives of the product solution.

$$u(x, y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = v'(x)w(y)$$

$$\frac{\partial u}{\partial y} = v(x)w'(y) \qquad \frac{\partial^2 u}{\partial y^2} = v(x)w''(y)$$

Then we'll plug the derivatives into the partial differential equation.

$$3v'(x)w(y) - v(x)w''(y) = 0$$

$$3v'(x)w(y) = v(x)w''(y)$$

$$3\frac{dv}{dx}w(y) = v(x)\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v(x)}\right)\frac{dv}{dx} = \left(\frac{1}{3w(y)}\right)\frac{d^2w}{dy^2}$$

The equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. The only way this can possibly be true is if both functions are constant, so we'll set the equation equal to $-\lambda$.

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$$\left(\frac{3}{v(x)}\right)\frac{dv}{dx} = \left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{v(x)}\right)\frac{dv}{dx} = -\lambda$$

$$\ln v = -\lambda x + C$$

$$v = Ce^{-\lambda x}$$

and

$$\left(\frac{1}{3w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} + 3\lambda w = 0$$

$$r^2 + 3\lambda = 0$$

$$r = \pm\sqrt{-3\lambda}$$

$$w(y) = c_1\cos(\sqrt{3\lambda}y) + c_2\sin(\sqrt{3\lambda}y)$$

Then

$$u(x,y) = Ce^{-\lambda x}(c_1\cos(\sqrt{3\lambda}y) + c_2\sin(\sqrt{3\lambda}y))$$

$$u(x,y) = c_1e^{-\lambda x}\cos(\sqrt{3\lambda}y) + c_2e^{-\lambda x}\sin(\sqrt{3\lambda}y)$$

Substituting $u = (0,y) = 4\sin(\sqrt{3\lambda}y)$ into the general solution gives

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$$c_1 \cos(\sqrt{3\lambda}y) + c_2 \sin(\sqrt{3\lambda}y) = 4 \sin(\sqrt{3\lambda}y)$$

We'll have $c_1 = 0$ and $c_2 = 4$, which means the solution to the boundary value problem is

$$u(x, y) = 4e^{-\lambda x} \sin(\sqrt{3\lambda}y)$$

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THE HEAT EQUATION

■ 1. Find a solution to the partial differential equation if $u(x,0) = \sin(\pi x) + 3\sin(\pi x/2)$ and $u(0,t) = u(6,t) = 0$.

$$\frac{\partial u}{\partial t} = 9\frac{\partial^2 u}{\partial x^2} \qquad 0 \leq x \leq 6$$

Solution:

If we start with the product solution $u(x,t) = v(x)w(t)$, then we know

$$v(x)w'(t) = 9v''(x)w(t)$$

$$\frac{w'(t)}{9w(t)} = \frac{v''(x)}{v(x)} = -\lambda$$

Break the equation into two ordinary differential equations.

$$w' = -9\lambda w$$

$$v'' = -\lambda v$$

The solution to the first equation, which is the first order differential equation, is

$$w(t) = Ce^{-9\lambda t}$$

To solve the second equation, we find the associated characteristic equation

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$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$ and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions $v(0) = 0$ and $v(6) = 0$ gives

$$c_1 + c_2 = 0$$

$$c_1 e^{-6\sqrt{-\lambda}} + c_2 e^{6\sqrt{-\lambda}} = 0$$

which allows use to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x, t) = 0$.

If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$, and the solution is

$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v(6) = c_1 + 6c_2 = 0$$

which allows us to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x, t) = 0$.

If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$, and the solution is

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$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives us

$$v(0) = c_1 = 0$$

$$v(6) = c_1 \cos(6\sqrt{\lambda}) + c_2 \sin(6\sqrt{\lambda}) = 0$$

$$c_2 \sin(6\sqrt{\lambda}) = 0$$

We'll find a non-trivial solution which satisfies

$$\sin(6\sqrt{\lambda}) = 0$$

$$6\sqrt{\lambda} = n\pi \qquad\qquad n = 1, 2, 3, \ldots$$

$$\lambda_n = \left(\frac{\pi n}{6}\right)^2$$

So the general solution for $\lambda > 0$ becomes

$$v(x) = c_2 \sin\left(\sqrt{\left(\frac{\pi n}{6}\right)^2}\, x\right)$$

$$v(x) = c_2 \sin\left(\frac{n\pi x}{6}\right) \qquad n = 1, 2, 3...$$

$$w(t) = Ce^{-9\lambda t}$$

$$w(t) = Ce^{-9\left(\frac{\pi n}{6}\right)^2 t}$$

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Putting the results together from both equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u(x,t) = c_2 \sin\left(\frac{n\pi x}{6}\right) Ce^{-9\left(\frac{n\pi}{6}\right)^2 t} \qquad n = 1,\ 2,\ 3...$$

While c_2 will be different for each n, the constant $C \times c_2$ will also depend on n, so let's rename it as B_n.

$$u_n(x,t) = \sum_{n=1}^{\infty} B_n \sin\frac{\pi n x}{6} e^{-9\left(\frac{\pi n}{6}\right)^2 t}$$

Matching up the boundary conditions gives us

$$u(x,0) = \sin(\pi x) + 3\sin(\pi x/2)$$

and then

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{6}\right)(1) = \sin(\pi x) + 3\sin\left(\frac{\pi x}{2}\right)$$

We get $B_6 = 1$, $B_3 = 3$, with all other values of $B_n = 0$, so the solution is

$$u(x,t) = e^{-9\pi^2 t}\sin(\pi x) + 3e^{-\frac{9}{4}\pi^2 t}\sin\left(\frac{\pi x}{2}\right)$$

■ 2. Find a solution to the partial differential equation if $u(x,0) = 2\cos(\pi x/2L)$, $u_x(0,t) = 0$, and $u(L,t) = 0$.

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$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \qquad 0 \leq x \leq L$$

Solution:

We start with the product solution $u(x,t) = v(x)w(t)$, then

$$v(x)w'(t) = 4v''(x)w(t)$$

$$\frac{w'(t)}{4w(t)} = \frac{v''(x)}{v(x)} = -\lambda$$

Break the equation into two ordinary differential equations.

$$w' = -4\lambda w$$

$$v'' = -\lambda v$$

The solution to the first equation, which is the first order differential equation, is

$$w(t) = Ce^{-4\lambda t}$$

To solve the second equation, we find the associated characteristic equation

$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$, and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

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Applying the boundary conditions gives us

$$v'(0) = -c_1\sqrt{-\lambda} + c_2\sqrt{-\lambda} = 0 \qquad c_1 = c_2$$

$$v(L) = c_1 e^{-L\sqrt{-\lambda}} + c_2 e^{L\sqrt{-\lambda}} = 0$$

which allows use to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x, t) = 0$.

If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$, and the solution is

$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives us

$$v'(0) = c_2 = 0$$

$$v(L) = c_1 + c_2 L = 0$$

which allows us to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x, t) = 0$.

If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$, and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives us

$$v'(0) = \sqrt{\lambda}c_2 = 0 \qquad c_2 = 0$$

$$v(L) = c_1 \cos(\sqrt{\lambda}L) + c_2 \sin(\sqrt{\lambda}L) = c_1 \cos(\sqrt{\lambda}L) = 0$$

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We'll find a non-trivial solution which satisfies

$$\cos(\sqrt{\lambda}L) = 0$$

$$\sqrt{\lambda}L = \frac{\pi}{2} + n\pi$$

$$\lambda_n = \left(\frac{\pi + 2n\pi}{2L}\right)^2$$

So the general solution for $\lambda > 0$ becomes

$$v(x) = c_1 \cos\frac{(\pi + 2n\pi)x}{2L}$$

$$w(t) = Ce^{-4\lambda t}$$

$$w(t) = Ce^{-4\left(\frac{\pi + 2n\pi}{2L}\right)^2 t} = Ce^{-\left(\frac{\pi(2n+1)}{L}\right)^2 t} \qquad n = 1, 2, 3...$$

Putting the results together from both equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u(x,t) = c_1 \cos\left(\frac{(\pi + 2n\pi)x}{2L}\right) Ce^{-\left(\frac{\pi(2n+1)}{L}\right)^2 t}$$

While c_1 will be different for each n, the constant $C \times c_1$ will also depend on n, so let's rename it as A_n.

$$u_n(x,t) = A_n \cos\left(\frac{(\pi + 2n\pi)x}{2L}\right) e^{-\left(\frac{\pi(2n+1)}{L}\right)^2 t} \qquad n = 1, 2, 3...$$

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The solution is

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{(\pi + 2n\pi)x}{2L}\right) e^{-\left(\frac{\pi(2n+1)}{L}\right)^2 t}$$

Matching up the boundary conditions gives us

$$u(x,0) = 2\cos\left(\frac{\pi x}{2L}\right)$$

and then

$$\sum_{n=1}^{\infty} A_n \cos\left(\frac{(\pi + 2n\pi)x}{2L}\right) = 2\cos\left(\frac{\pi x}{2L}\right)$$

We get $A_0 = 2$, but then $A_1 = A_2 = A_3 = \ldots = 0$, so the solution is

$$u(x,t) = 2\cos\left(\frac{\pi x}{2L}\right) e^{-\frac{\pi^2}{L^2}t}$$

■ 3. Find a solution to the partial differential equation if $u(x,0) = 4\sin(\pi x/6)\cos^2(\pi x/6)$, $u(0,t) = 0$, and $u_x(3,t) = 0$.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 3$$

Solution:

We start with the product solution $u(x,t) = v(x)w(t)$, then

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$$v(x)w'(t) = v''(x)w(t)$$

$$\frac{w'(t)}{w(t)} = \frac{v''(x)}{v(x)} = -\lambda$$

Break the equation into two ordinary differential equations

$$w' = -\lambda w$$

$$v'' = -\lambda v$$

The solution to the first equation, which is the first order differential equation, is

$$w(t) = Ce^{-\lambda t}$$

To solve the second equation, we find the associated characteristic equation

$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$, and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions gives

$$v(0) = c_1 + c_2 = 0 \qquad c_1 = c_2$$

$$v'(3) = -3\sqrt{-\lambda}c_1 e^{-3\sqrt{-\lambda}} + 3\sqrt{-\lambda}c_2 e^{3\sqrt{-\lambda}} = 0$$

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which allows us to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x,t) = 0$.

If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$, and the solution is

$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v(3) = c_2 = 0$$

which allows us to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x,t) = 0$.

If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$, and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v'(3) = -c_1 \sin(3\sqrt{\lambda}) + c_2 \cos(3\sqrt{\lambda}) = c_2 \cos(3\sqrt{\lambda}) = 0$$

We'll find a non-trivial solution which satisfies

$$\cos(3\sqrt{\lambda}) = 0$$

$$3\sqrt{\lambda} = \frac{\pi}{2} + n\pi$$

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$$\lambda_n = \left(\frac{\pi + 2n\pi}{6}\right)^2$$

So the general solution for $\lambda > 0$ becomes

$$v(x) = c_2 \sin \frac{(\pi + 2n\pi)x}{6}$$

$$w(t) = Ce^{-\lambda t}$$

$$w(t) = Ce^{-\left(\frac{\pi + 2n\pi}{6}\right)^2 t} \qquad n = 1, 2, 3...$$

Putting the results together from both equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u(x,t) = c_2 \sin \frac{(\pi + 2n\pi)x}{6} Ce^{-\left(\frac{\pi(2n+1)}{6}\right)^2 t}$$

While c_2 will be different for each n, the constant $C \times c_2$ will also depend on n, so let's rename it as B_n.

$$u_n(x,t) = B_n \sin\left(\frac{(\pi + 2n\pi)x}{6}\right) e^{-\left(\frac{\pi(2n+1)}{6}\right)^2 t}, \quad n = 1, 2, 3...$$

The solution is

$$u(x,t) = \sum_{n=0}^{\infty} B_n \sin\left(\frac{(\pi + 2n\pi)x}{6}\right) e^{-\left(\frac{\pi(2n+1)}{6}\right)^2 t}$$

Matching up the boundary conditions gives

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$$u(x,0) = 4\sin\left(\frac{\pi x}{6}\right)\cos^2\left(\frac{\pi x}{6}\right)$$

Then we get

$$\sum_{n=0}^{\infty} B_n \sin\left(\frac{(\pi + 2n\pi)x}{6}\right) = 4\sin\left(\frac{\pi x}{6}\right)\cos^2\left(\frac{\pi x}{6}\right)$$

$$\sum_{n=0}^{\infty} B_n \sin\left(\frac{(\pi + 2n\pi)x}{6}\right) = 2\sin\left(\frac{\pi x}{3}\right)\cos\left(\frac{\pi x}{6}\right)$$

$$\sum_{n=0}^{\infty} B_n \sin\left(\frac{(\pi + 2n\pi)x}{6}\right) = \sin\left(\frac{\pi x}{6}\right) + \sin\left(\frac{\pi x}{2}\right)$$

We find $B_0 = 1$, $B_1 = 1$, and $B_2 = B_3 = \ldots = 0$, so the solution is

$$u(x,t) = \sin\left(\frac{\pi x}{6}\right)e^{-\frac{\pi^2}{36}t} + \sin\left(\frac{\pi x}{2}\right)e^{-\frac{\pi^2}{4}t}$$

■ 4. Find a solution to the partial differential equation if $u(x,0) = 2x + 5$, $u_x(0,t) = 0$, and $u_x(L,t) = 0$.

$$\frac{\partial u}{\partial t} = 25\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le L$$

Solution:

We start with the product solution $u(x,t) = v(x)w(t)$, then

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$$v(x)w'(t) = 25v''(x)w(t)$$

$$\frac{w'(t)}{25w(t)} = \frac{v''(x)}{v(x)} = -\lambda$$

Break the equation into two ordinary differential equations.

$$w' = -25\lambda w$$

$$v'' = -\lambda v$$

The solution to the first equation, which is the first order differential equation, is

$$w(t) = Ce^{-25\lambda t}$$

To solve the second equation, we find the associated characteristic equation

$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$, and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions gives

$$v'(0) = -c_1\sqrt{-\lambda} + c_2\sqrt{-\lambda} = 0$$

$$v'(L) = -c_1\sqrt{-\lambda}e^{-L\sqrt{-\lambda}} + c_2\sqrt{-\lambda}e^{-L\sqrt{-\lambda}} = 0$$

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which allows us to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x, t) = 0$.

If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$, and the solution is

$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives

$$v'(0) = c_2 = 0$$

$$v'(L) = c_2 = 0$$

so there's non-trivial solution $v(x) = C_1$ and $w(t) = C_2$.

If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$, and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives us

$$v'(0) = \sqrt{\lambda}c_2 = 0 \qquad\qquad c_2 = 0$$

$$v'(L) = -c_1\sqrt{\lambda}\sin(\sqrt{\lambda}L) + c_2\sqrt{\lambda}\cos(\sqrt{\lambda}L) = -c_1\sqrt{\lambda}\sin(\sqrt{\lambda}L) = 0$$

We'll find a non-trivial solution which satisfies

$$\sin(\sqrt{\lambda}L) = 0$$

$$\sqrt{\lambda}L = n\pi$$

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$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

So the general solution for $\lambda > 0$ becomes

$$v(x) = c_1 \cos\left(\frac{n\pi x}{L}\right)$$

$$w(t) = Ce^{-25\lambda t}$$

$$w(t) = Ce^{-25\left(\frac{\pi n}{L}\right)^2 t} \qquad n = 1, 2, 3...$$

Putting the results together from both equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u_n(x,t) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-25\left(\frac{n\pi}{L}\right)^2 t}$$

Matching up the boundary conditions gives

$$u(x,0) = 2x + 5$$

Then we get

$$A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-25\left(\frac{n\pi}{L}\right)^2 t} = 2x + 5$$

We use the formulas for A_0 and A_n from the Fourier series representation to get

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$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^L 2x + 5\ dx = \frac{1}{L}(x^2 + 5x)\Big|_0^L = L + 5$$

and

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{L}\int_0^L (2x + 5)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

Use integration by parts, letting $u = 2x + 5$, $du = 2\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2}{n\pi}(2x + 5)\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L - \frac{4}{n\pi}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = -\frac{4}{n\pi}\left(-\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L$$

$$A_n = \frac{4L}{(n\pi)^2}((-1)^n - 1)$$

So the solution is

$$u(x,t) = 5 + L + \sum_{n=1}^{\infty}\frac{4L}{(n\pi)^2}((-1)^n - 1)\cos\left(\frac{n\pi x}{L}\right)e^{-25\left(\frac{n\pi}{L}\right)^2 t}$$

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■ 5. Find a solution to the partial differential equation if $u(x,0) = x \sin x$, $u(0,t) = 0$, and $u(\pi, t) = 0$.

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \qquad 0 \leq x \leq \pi$$

Solution:

We start with the product solution $u(x, t) = v(x)w(t)$, then

$$v(x)w'(t) = 4v''(x)w(t)$$

$$\frac{w'(t)}{4w(t)} = \frac{v''(x)}{v(x)} = -\lambda$$

Break the equation into two ordinary differential equations.

$$w' = -4\lambda w$$

$$v'' = -\lambda v$$

The solution to the first equation, which is the first order differential equation, is

$$w(t) = Ce^{-4\lambda t}$$

To solve the second equation, we find the associated characteristic equation.

$$r^2 + \lambda = 0$$

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From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$, and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions gives

$$v(0) = c_1 + c_2 = 0 \qquad\qquad c_1 = c_2$$

$$v(\pi) = c_1 e^{-\pi\sqrt{-\lambda}} + c_2 e^{\pi\sqrt{-\lambda}} = 0$$

which allows us to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x, t) = 0$.

If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$, and the solution is

$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives us

$$v(0) = c_1 = 0$$

$$v(\pi) = c_1 + c_2\pi = 0$$

which allows us to determine that $c_1 = c_2 = 0$, from which we can get the trivial solution only $u(x, t) = 0$.

If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$, and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

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Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v(\pi) = c_1 \cos(\pi\sqrt{\lambda}) + c_2 \sin(\pi\sqrt{\lambda}) = c_2 \sin(\pi\sqrt{\lambda}) = 0$$

We'll find a non-trivial solution which satisfies

$$\sin(\pi\sqrt{\lambda}) = 0$$

$$\pi\sqrt{\lambda} = n\pi$$

$$\lambda_n = n^2$$

So the general solution for $\lambda > 0$ becomes

$$v(x) = c_2 \sin(nx)$$

$$w(t) = Ce^{-4\lambda t}$$

$$w(t) = Ce^{-4n^2 t} \qquad n = 1, 2, 3...$$

Putting the results together from both equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u(x,t) = c_2 \sin(nx)Ce^{-4n^2 t}$$

While c_2 will be different for each n, the constant $C \times c_2$ will also depend on n, so let's rename it as B_n.

$$u_n(x,t) = B_n \sin(nx)e^{-4n^2 t} \qquad n = 1, 2, 3...$$

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The solution is

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin(nx) e^{-4n^2 t}$$

Matching up the boundary conditions gives

$$u(x,0) = x \sin x$$

Then we get

$$\sum_{n=1}^{\infty} B_n \sin(nx) e^{-4n^2 t} = x \sin x$$

We use the formula for B_n from the Fourier series representation to get

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$B_n = \frac{2}{\pi} \int_0^{\pi} x \sin x \sin(nx) \, dx$$

$$B_n = \frac{1}{\pi} \int_0^{\pi} x(\cos(n-1)x - \cos(n+1)x) \, dx$$

Use integration by parts, letting $u = x$, $du = dx$, $dv = \cos(n-1)x - \cos(n+1)x \; dx$, and

$$v = -\frac{1}{n-1} \sin(n-1)x + \frac{1}{n+1} \sin(n+1)x$$

and we'll get

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$$B_n = \frac{1}{\pi}\left(x\left(-\frac{1}{n-1}\sin(n-1)x + \frac{1}{n+1}\sin(n+1)x\right)\Big|_0^{\pi} \right.$$

$$\left. -\int_0^{\pi}\left(-\frac{1}{n-1}\sin(n-1)x + \frac{1}{n+1}\sin(n+1)x\right)dx \right)$$

$$B_n = \frac{1}{\pi}\left(-\frac{1}{(n-1)^2}\cos(n-1)x + \frac{1}{(n+1)^2}\cos(n+1)x\right)\Big|_0^{\pi}$$

$$B_n = \frac{1}{\pi}\left(\frac{1}{(n+1)^2}((-1)^{n+1}-1) - \frac{1}{(n-1)^2}((-1)^{n-1}-1)\right)$$

$$B_n = -\frac{4n((-1)^n+1)}{\pi(n^2-1)^2},\ n \neq 1$$

For $n = 1$ specifically, we find

$$B_1 = \frac{2}{\pi}\int_0^{\pi} x\sin^2 x\ dx$$

$$B_1 = \frac{1}{\pi}\int_0^{\pi} x(1-\cos 2x)\ dx$$

$$B_1 = -\frac{1}{\pi}\left(\frac{x^2}{2}\right)\Big|_0^{\pi} - \frac{1}{\pi}\int_0^{\pi} x\cos(2x)\ dx$$

Use integration by parts, letting $u = x$, $du = dx$, $dv = \cos(2x)\ dx$, and $v = (1/2)\sin(2x)$.

$$B_1 = \frac{\pi}{2} - \frac{1}{\pi}\left(\frac{x}{2}\sin(2x)\Big|_0^{\pi} - \frac{1}{2}\int_0^{\pi}\sin(2x)\ dx\right)$$

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$$B_1 = \frac{\pi}{2} + \frac{1}{2\pi}\left(-\frac{1}{2}\cos(2x)\Big|_0^{\pi}\right)$$

$$B_1 = \frac{\pi}{2}$$

Then the solution is

$$u(x,t) = \frac{\pi}{2}e^{-4t}\sin x - \sum_{n=2}^{\infty}\frac{4n((-1)^n+1)}{\pi(n^2-1)^2}\sin(nx)e^{-4n^2t}$$

■ 6. Find a solution to the partial differential equation if $u(x,0) = \sin(\pi x)$, $u_x(0,t) = 0$, and $u_x(5,t) = 0$.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 5$$

Solution:

We start with the product solution $u(x, t) = v(x)w(t)$, then

$$v(x)w'(t) = v''(x)w(t)$$

$$\frac{w'(t)}{w(t)} = \frac{v''(x)}{v(x)} = -\lambda$$

Break the equation into two ordinary differential equations.

$$w' = -\lambda w$$

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$$v'' = -\lambda v$$

The solution to the first equation, which is the first order differential equation, is

$$w(t) = Ce^{-\lambda t}$$

To solve the second equation, we find the associated characteristic equation.

$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$, and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions gives

$$v'(0) = -c_1\sqrt{-\lambda} + c_2\sqrt{-\lambda} = 0 \qquad\qquad c_1 = c_2$$

$$v'(5) = -c_1\sqrt{-\lambda}e^{-5\sqrt{-\lambda}} + c_2\sqrt{-\lambda}e^{5\sqrt{-\lambda}} = 0$$

which allows use to determine that $c_1 = c_2 = 0$, from which we get only the trivial solution $u(x,t) = 0$.

If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$, and the solution is

$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives

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$$v'(0) = c_2 = 0$$

$$v'(5) = c_2 = 0$$

so the solution is $v_0(x) = c_1$ and $w_0(t) = C$.

If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$, and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives

$$v'(0) = \sqrt{\lambda}c_2 = 0 \qquad\qquad c_2 = 0$$

$$v'(5) = -c_1\sqrt{\lambda}\sin(5\sqrt{\lambda}) + c_2\cos\sqrt{\lambda}(5\sqrt{\lambda}) = -c_1\sqrt{\lambda}\sin(5\sqrt{\lambda}) = 0$$

We'll find a non-trivial solution which satisfies

$$\sin(5\sqrt{\lambda}) = 0$$

$$5\sqrt{\lambda} = n\pi$$

$$\lambda_n = \left(\frac{n\pi}{5}\right)^2$$

So the general solution for $\lambda > 0$ becomes

$$v(x) = c_1 \cos\left(\frac{n\pi x}{5}\right)$$

$$w(t) = Ce^{-\lambda t}$$

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$$w(t) = Ce^{-\left(\frac{n\pi}{5}\right)^2 t} \qquad n = 1, 2, 3...$$

Putting the results together from both equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u(x,t) = c_1 C + c_1 \cos\left(\frac{n\pi x}{5}\right) Ce^{-\left(\frac{n\pi}{5}\right)^2 t}$$

While c_1 will be different for each n, the constant $C \times c_1$ will also depend on n, so let's rename it as A_n.

$$u_n(x,t) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{5}\right) e^{-\left(\frac{n\pi}{5}\right)^2 t} \qquad n = 1, 2, 3...$$

Matching up the boundary conditions gives

$$u(x,0) = \sin(\pi x)$$

We use the formulas for A_0 and A_n from the Fourier series representation to get

$$A_0 = \frac{1}{L}\int_0^L f(x)\, dx$$

$$A_0 = \frac{1}{5}\int_0^5 \sin(\pi x)\, dx = \frac{1}{5}\left(-\frac{1}{\pi}\right)\cos(\pi x)\Big|_0^5 = \frac{2}{5\pi}$$

and

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$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{5}\int_0^5 \sin(\pi x)\cos\left(\frac{n\pi x}{5}\right)\,dx$$

$$A_n = \frac{1}{5}\int_0^5 \left(\sin\pi x\left(1-\frac{n}{5}\right)+\sin\pi x\left(1+\frac{n}{5}\right)\right)dx$$

$$A_n = -\frac{1}{5}\left(\frac{1}{\pi\left(1-\frac{n}{5}\right)}\cos\left(\pi+\frac{n\pi}{5}\right)x+\frac{1}{\pi\left(1+\frac{n}{5}\right)}\cos\left(\pi+\frac{n\pi}{5}\right)x\right)\Bigg|_0^5$$

$$A_n = -\frac{10(1+(-1)^n)}{\pi(n^2-25)},\ n\neq 5$$

For $n = 5$ specifically, we find

$$A_5 = \frac{2}{5}\int_0^5 \sin(\pi x)\cos(\pi x)\,dx$$

$$A_5 = \frac{1}{5}\int_0^5 \sin(2\pi x)\,dx = \frac{1}{10\pi}\cos(2\pi x)\Big|_0^5 = 0$$

so the solution is

$$u(x,t) = \frac{2}{5\pi} + \sum_{n=1,n\neq 5}^{\infty} -\frac{10(1+(-1)^n)}{\pi(n^2-25)}\cos\left(\frac{n\pi x}{5}\right)e^{-\left(\frac{n\pi}{5}\right)^2 t}$$

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CHANGING THE TEMPERATURE BOUNDARIES

■ 1. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2} \qquad 0 \leq x \leq \pi$$

$$u(x,0) = 12x + \pi$$

$$u(0,t) = \pi \qquad u(\pi, t) = 3\pi$$

Solution:

Find the function that models equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E = \pi + \frac{3\pi - \pi}{\pi}x$$

$$u_E = 2x + \pi$$

Find the coefficients B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{\pi}\int_0^\pi (12x + \pi - 2x - \pi)\sin(nx)\,dx$$

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$$B_n = \frac{2}{\pi}\int_0^{\pi} 10x\sin(nx)\,dx$$

Integrate, using integration by parts for the second integral with $u = x$, $du = dx$, $dv = \sin(nx)\,dx$, and $v = -(1/n)\cos(nx)$.

$$B_n = \frac{20}{\pi}\left(x\left(-\frac{1}{n}\cos(nx)\right)\Big|_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos(nx)\,dx\right)$$

$$B_n = \frac{20}{\pi}\left(-\frac{\pi}{n}(-1)^n + \frac{1}{n^2}\sin(nx)\Big|_0^{\pi}\right) = \frac{20}{n}(-1)^{n+1}$$

Then the solution of the equation is

$$u(x,t) = u_E + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = 2x + \pi + \sum_{n=1}^{\infty}\frac{20(-1)^{n+1}}{n}\sin(nx)\,e^{-2n^2 t}$$

2. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 4$$

$$u(x,0) = x^2 + 1$$

$$u(0,t) = 1 \qquad u(4,t) = 17$$

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Solution:

Find the function that models equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E = 1 + \frac{17 - 1}{4}x$$

$$u_E = 1 + 4x$$

Find the coefficients B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{2}\int_0^4 (x^2 + 1 - 1 - 4x)\sin\left(\frac{n\pi x}{4}\right)\,dx$$

$$B_n = \frac{1}{2}\int_0^4 (x^2 - 4x)\sin\left(\frac{n\pi x}{4}\right)\,dx$$

Integrate, using integration by parts for the second integral with $u = x^2 - 4x$, $du = (2x - 4)\,dx$, $dv = \sin(n\pi x/4)\,dx$, and $v = -(4/n\pi)\cos(n\pi x/4)$.

$$B_n = \frac{1}{2}\left[(x^2 - 4x)\left(-\frac{4}{n\pi}\cos\left(\frac{n\pi x}{4}\right)\right)\Bigg|_0^4 + \frac{4}{n\pi}\int_0^4 (2x - 4)\cos\left(\frac{n\pi x}{4}\right)\,dx\right]$$

$$B_n = \frac{2}{n\pi}\int_0^4 (2x - 4)\cos\left(\frac{n\pi x}{4}\right)\,dx$$

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Integrate, using integration by parts for the second integral with $u = 2x - 4$, $du = 2\ dx$, $dv = \cos(n\pi x/4)\ dx$, and $v = (4/n\pi)\sin(n\pi x/4)$.

$$B_n = \frac{2}{n\pi}\left[(2x-4)\frac{4}{n\pi}\sin\left(\frac{n\pi x}{4}\right)\Big|_0^4 - \frac{8}{n\pi}\int_0^4 \sin\left(\frac{n\pi x}{4}\right)\ dx\right]$$

$$B_n = \frac{16}{n^2\pi^2}\cdot\frac{4}{n\pi}\cos\left(\frac{n\pi x}{4}\right)\Big|_0^4 = \frac{64}{n^3\pi^3}((-1)^n - 1)$$

Then the solution of the equation is

$$u(x,t) = u_E + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = 1 + 4x + \sum_{n=1}^{\infty}\frac{64}{n^3\pi^3}((-1)^n - 1)\sin\left(\frac{n\pi x}{4}\right)e^{-\frac{n^2\pi^2}{4}t}$$

■ 3. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 3\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 1$$

$$u(x,0) = x + \sin(\pi x)$$

$$u(0,t) = 4 \qquad u(1,t) = 5$$

Solution:

Find the function that models equilibrium temperature.

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$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E = 4 + \frac{5-4}{1}x$$

$$u_E = x + 4$$

Find the coefficients B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = 2\int_0^1 (x + \sin(\pi x) - x - 4)\sin(n\pi x)\,dx$$

$$B_n = 2\int_0^1 \sin(\pi x)\sin(n\pi x) - 4\sin(n\pi x)\,dx$$

$$B_n = \int_0^1 \cos(n-1)\pi x - \cos(n+1)\pi x - 8\sin(n\pi x)\,dx$$

$$B_n = \frac{1}{\pi(n-1)}\sin(n-1)\pi x - \frac{1}{\pi(n+1)}\sin(n+1)\pi x + \frac{8}{n\pi}\cos(n\pi x)\Big|_0^1$$

$$B_n = \frac{8}{n\pi}((-1)^n - 1),\ n \neq 1$$

For $n = 1$ specifically, we get

$$B_1 = 2\int_0^1 (\sin\pi x - 4)\sin(\pi x)\,dx$$

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$$B_1 = \int_0^1 1 - \cos(2\pi x) - 8\sin(\pi x)\ dx$$

$$B_1 = \left(x - \frac{1}{2\pi}\sin(2\pi x) + \frac{8}{\pi}\cos(\pi x) \right)\Bigg|_0^1 = 1 - \frac{16}{\pi}$$

Then the solution of the equation is

$$u(x,t) = u_E + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = x + 4 + \left(1 - \frac{16}{\pi}\right)\sin(\pi x)e^{-3\pi^2 t} + \sum_{n=2}^{\infty} \frac{8((-1)^n - 1)}{\pi n}\sin(n\pi x)e^{-3\pi^2 n^2 t}$$

4. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 5\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 10$$

$$u(x,0) = 6 - 2x$$

$$u(0,t) = 10 \qquad u(10,t) = 50$$

Solution:

Find the function that models equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

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$$u_E = 10 + \frac{50 - 10}{10}x$$

$$u_E = 10 + 4x$$

Find the coefficients B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{5}\int_0^{10} (6 - 2x - 10 - 4x)\sin\left(\frac{n\pi x}{10}\right)\,dx$$

$$B_n = -\frac{1}{5}\int_0^{10} (4 + 6x)\sin\left(\frac{n\pi x}{10}\right)\,dx$$

Integrate, using integration by parts for the second integral with $u = 6x + 4$, $du = 6\,dx$, $dv = \sin(n\pi x/10)\,dx$, and $v = -(10/n\pi)\cos(n\pi x/10)$.

$$B_n = -\frac{1}{5}\left(-\frac{10}{n\pi}(6x + 4)\cos\left(\frac{n\pi x}{10}\right)\Big|_0^{10} + \frac{60}{n\pi}\int_0^{10}\cos\left(\frac{n\pi x}{10}\right)\,dx\right)$$

$$B_n = \frac{2}{n\pi}(64(-1)^n - 4) - \frac{12}{n\pi}\left(\frac{10}{n\pi}\right)\sin\left(\frac{n\pi x}{10}\right)\Big|_0^{10}$$

$$B_n = -\frac{8}{n\pi} + \frac{128(-1)^n}{n\pi} = \frac{8}{n\pi}(16(-1)^n - 1)$$

Then the solution of the equation is

$$u(x,t) = u_E + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

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$$u(x,t) = 10 + 4x + \sum_{n=1}^{\infty} \frac{8}{n\pi}(16(-1)^n - 1)\sin\left(\frac{n\pi x}{10}\right) e^{-5\left(\frac{n\pi}{10}\right)^2 t}$$

■ 5. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \qquad 0 \leq x \leq \frac{\pi}{2}$$

$$u(x,0) = \cos x$$

$$u(0,t) = 1 \qquad u\left(\frac{\pi}{2}, t\right) = 0$$

Solution:

Find the function that models equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E = 1 + \frac{0-1}{\frac{\pi}{2}}x$$

$$u_E = 1 - \frac{2}{\pi}x$$

Find the coefficients B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{4}{\pi}\int_0^{\frac{\pi}{2}} \left(\cos x + \frac{2}{\pi}x - 1\right)\sin(2nx)\ dx$$

$$B_n = \frac{4}{\pi}\left[\int_0^{\frac{\pi}{2}} \frac{2}{\pi}x\sin(2nx)\ dx\right.$$

$$\left. + \int_0^{\frac{\pi}{2}} \frac{1}{2}\sin(2n-1)x + \frac{1}{2}\sin(2n+1)x - \sin(2nx)\ dx\right]$$

Integrate, using integration by parts for the first integral with $u = x$, $du = \ dx$, $dv = \sin(2nx)\ dx$, and $v = -(1/2n)\cos(2nx)$.

$$B_n = \frac{8}{\pi^2}\left(-\frac{1}{2n}x\cos(2nx)\Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \frac{1}{2n}\cos(2nx)\ dx\right)$$

$$-\frac{2}{\pi(2n-1)}\cos(2n-1)x\Big|_0^{\frac{\pi}{2}} - \frac{2}{\pi(2n+1)}\cos(2n+1)x\Big|_0^{\frac{\pi}{2}} + \frac{2}{n\pi}\cos(2nx)\Big|_0^{\frac{\pi}{2}}$$

$$B_n = -\frac{4}{n\pi^2}\left((-1)^n\frac{\pi}{2}\right) + \frac{2}{n^2\pi^2}\sin(2nx)\Big|_0^{\frac{\pi}{2}}$$

$$+\frac{2}{\pi(2n-1)} + \frac{2}{\pi(2n+1)} + \frac{2}{n\pi}((-1)^n - 1)$$

$$B_n = \frac{2(n(2n+1) + n(2n-1) - 4n^2 + 1)}{n\pi(4n^2-1)} = \frac{2}{n\pi(4n^2-1)}$$

Then the solution of the equation is

$$u(x,t) = u_E + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

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$$u(x,t) = 1 - \frac{2}{\pi}x + \sum_{n=1}^{\infty} \frac{2}{n\pi(4n^2-1)} \sin(2nx)\ e^{-4n^2t}$$

6. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le \pi$$

$$u(x,0) = e^x + x + 1$$

$$u(0,t) = 1 \qquad u(\pi,t) = 1 + \pi$$

Solution:

Find the function that models equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E = 1 + \frac{1 + \pi - 1}{\pi}x$$

$$u_E = x + 1$$

Find the coefficients B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{2}{\pi}\int_0^{\pi} (e^x + x + 1 - x - 1)\sin(nx)\ dx$$

$$B_n = \frac{2}{\pi}\int_0^{\pi} e^x \sin(nx)\ dx$$

Integrate, using integration by parts for the second integral with $u = e^x$, $du = e^x\ dx$, $dv = \sin(nx)\ dx$, and $v = -(1/n)\cos(nx)$.

$$B_n = \frac{2}{\pi}\left(-\frac{1}{n}e^x \cos(nx)\Big|_0^{\pi} + \frac{1}{n}\int_0^{\pi} e^x \cos(nx)\ dx\right)$$

Integrate, using integration by parts for the second integral with $u = e^x$, $du = e^x\ dx$, $dv = \cos(nx)\ dx$, and $v = (1/n)\sin(nx)$.

$$B_n = \frac{2}{\pi}\left[-\frac{1}{n}(e^{\pi}(-1)^n - 1) + \frac{1}{n}\left(\frac{1}{n}e^x \sin(nx)\Big|_0^{\pi} - \frac{1}{n}\int_0^{\pi} e^x \sin(nx)\ dx\right)\right]$$

$$B_n = -\frac{2}{n\pi}(e^{\pi}(-1)^n - 1) - \frac{2}{\pi n^2}\left(\frac{\pi}{2}\right)B_n$$

$$B_n\left(1 + \frac{1}{n^2}\right) = \frac{2}{n\pi}(1 - e^{\pi}(-1)^n)$$

$$B_n = \frac{2n(1 - e^{\pi}(-1)^n)}{\pi(n^2 + 1)}$$

Then the solution of the equation is

$$u(x,t) = u_E + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

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$$u(x,t) = 1 + x + \sum_{n=1}^{\infty} \frac{2n(1 - e^{\pi}(-1)^n)}{\pi(n^2+1)} \sin(nx)\ e^{-2n^2 t}$$

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LAPLACE'S EQUATION

■ 1. Use the product solution to find $u_1(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq \pi$ and $0 \leq y \leq 1$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 3\sin(2x) \qquad u(0,y) = 0$$

$$u(x,1) = 0 \qquad u(\pi, y) = 0$$

Solution:

If we find the first and second derivatives of the product solution,

$$u(x, y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2 v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2 w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2 v}{dx^2}w + v\frac{d^2 w}{dy^2} = 0$$

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then we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

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Our boundary conditions are $w(1) = 0$, $v(0) = 0$, and $v(\pi) = 0$, and the solution to the first ordinary differential equation boundary value problem is

$$\lambda_n = \left(\frac{n\pi}{\pi}\right)^2 = n^2 \qquad v_n(x) = C\sin(nx) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the second equation, we get

$$\frac{d^2w}{dy^2} - n^2 w = 0$$

$$w(1) = 0$$

$$w(y) = c_1 e^{-ny} + c_2 e^{ny}$$

Equivalently, this can be rewritten as

$$w(y) = c_1 \sinh n(y-1) + c_2 \cosh n(y-1)$$

which allows us to apply $w(1) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$w(y) = c_1 \sinh n(y-1)$$

Then the product solution is

$$u(x, y) = v(x)w(y)$$

$$u(x, y) = C\sin(nx)c_1 \sinh n(y-1)$$

$$u_n(x, y) = B_n \sin(nx)\sinh n(y-1) \qquad n = 1, 2, 3, \ldots$$

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$$u(x,y) = \sum_{n=1}^{\infty} B_n \sin(nx) \sinh n(y-1)$$

Substitute $u(x,0) = 3\sin(2x)$.

$$\sum_{n=1}^{\infty} B_n \sin(nx) \sinh n(y-1) = 3\sin(2x)$$

$$B_2 = -\frac{3}{\sinh 2}$$

Other than B_2, we get $B_1 = B_3 = \ldots = B_n = 0$. Then the solution is

$$u(x,y) = -\frac{3}{\sinh 2} \sin(2x) \sinh(2y-2)$$

■ 2. Use the product solution to find $u(x,y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \le x \le 3$ and $0 \le y \le 3$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u(0,y) = 0$$

$$u(x,3) = 2\sin(\pi x) + \sin(2\pi x) \qquad u(3,y) = 0$$

Solution:

If we find the first and second derivatives of the product solution,

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$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

then we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

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$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

Our boundary conditions are $w(0) = 0$, $v(0) = 0$, and $v(3) = 0$, and the solution to the first ordinary differential equation boundary value problem is

$$\lambda_n = \left(\frac{n\pi}{3}\right)^2 \qquad v_n(x) = C\sin(\sqrt{\lambda}x) = C\sin\left(\frac{n\pi x}{3}\right) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the second equation, we get

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

$$w(0) = 0$$

$$w(y) = c_1 e^{-\sqrt{\lambda}y} + c_2 e^{\sqrt{\lambda}y}$$

Equivalently, this can be rewritten as

$$w(y) = c_1\sinh(\sqrt{\lambda}y) + c_2\cosh(\sqrt{\lambda}y)$$

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which allows us to apply $w(0) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$w(y) = c_1 \sinh(\sqrt{\lambda} y)$$

$$w(y) = c_1 \sinh\left(\frac{n\pi y}{3}\right)$$

Then the product solution is

$$u(x, y) = v(x)w(y)$$

$$u(x, y) = C \sin\left(\frac{n\pi x}{3}\right) c_1 \sinh\left(\frac{n\pi y}{3}\right)$$

$$u_n(x, y) = B_n \sin\left(\frac{n\pi x}{3}\right) \sinh\left(\frac{n\pi y}{3}\right) \qquad n = 1, 2, 3, \ldots$$

$$u_n(x, y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{3}\right) \sinh\left(\frac{n\pi y}{3}\right)$$

Substitute $u(x,3) = 2\sin(\pi x) + \sin(2\pi x)$.

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{3}\right) \sinh(n\pi) = 2\sin(\pi x) + \sin(2\pi x)$$

$$B_3 = \frac{2}{\sinh(3\pi)}, \; B_6 = \frac{1}{\sinh(6\pi)}$$

Then the solution is

$$u(x, y) = \frac{2}{\sinh(3\pi)} \sin(\pi x)\sinh(\pi y) + \frac{1}{\sinh(6\pi)} \sin(2\pi x)\sinh(2\pi y)$$

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■ 3. Use the product solution to find $u(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq \pi$ and $0 \leq y \leq 2\pi$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u_y(x,0) = 0 \qquad u(0,y) = 2 + \cos(2y)$$

$$u_y(x,2\pi) = 0 \qquad u(\pi, y) = 0$$

Solution:

If we find the first and second derivatives of the product solution,

$$u(x, y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2 v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2 w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2 v}{dx^2}w + v\frac{d^2 w}{dy^2} = 0$$

then we can separate variables.

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$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \lambda$$

$$\frac{d^2v}{dx^2} = \lambda v(x)$$

$$\frac{d^2v}{dx^2} - \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

$$\frac{d^2w}{dy^2} = -\lambda w(y)$$

$$\frac{d^2w}{dy^2} + \lambda w = 0$$

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Our boundary conditions are $w(0) = 0$, $w(2\pi) = 0$, and $v(\pi) = 0$, and the solution to the second ordinary differential equation boundary value problem is

$$\lambda_0 = 0 \qquad\qquad w_0(y) = C$$

$$\lambda_n = \left(\frac{n\pi}{2\pi}\right)^2 = \frac{n^2}{4} \qquad w_n(x) = C\cos(\sqrt{\lambda}y) = C\cos\left(\frac{ny}{2}\right) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for $\lambda_0 = 0$ into the first equation is

$$v(x) = c_1 + c_2 x$$

and from $v(\pi) = 0$ we have

$$c_1 + c_2\pi = 0$$

$$v_0(x) = -c_2\pi + c_2 x = c_2(x - \pi)$$

When we plug this value for $\lambda_n > 0$ into the first equation, we get

$$v(x) = c_1 e^{-\sqrt{\lambda}x} + c_2 e^{\sqrt{\lambda}x}$$

Equivalently, this can be rewritten as

$$v(x) = c_1 \sinh\sqrt{\lambda}(x - \pi) + c_2 \cosh\sqrt{\lambda}(x - \pi)$$

which allows us to apply $v(\pi) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$v(x) = c_1 \sinh\frac{n(x - \pi)}{2}$$

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Then the product solution is

$$u(x, y) = v(x)w(y)$$

$$u(x, y) = c_2(x - \pi)C + c_1 \sinh\left(\frac{n(x - \pi)}{2}\right) C \cos\left(\frac{ny}{2}\right)$$

$$u_n(x, y) = A_0(x - \pi) + A_n \sinh\left(\frac{n(x - \pi)}{2}\right) \cos\left(\frac{ny}{2}\right) \qquad n = 1, 2, 3, \ldots$$

$$u_n(x, y) = A_0(x - \pi) + \sum_{n=1}^{\infty} A_n \sinh\left(\frac{n(x - \pi)}{2}\right) \cos\left(\frac{ny}{2}\right)$$

Substitute $u(0,y) = 2 + \cos(2y)$.

$$A_0(-\pi) + \sum_{n=1}^{\infty} A_n \sinh\left(-\frac{n\pi}{2}\right) \cos\left(\frac{ny}{2}\right) = 2 + \cos(2y)$$

$$A_0 = -\frac{2}{\pi}, \qquad A_4 = -\frac{1}{\sinh(2\pi)}$$

Then the solution is

$$u(x, y) = -\frac{2}{\pi}(x - \pi) - \frac{1}{\sinh 2\pi} \sinh(2x - 2\pi)\cos(2y)$$

■ 4. Use the product solution to find $u(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq 4$ and $0 \leq y \leq 6$.

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$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u_x(0,y) = 0$$

$$u_y(x,6) = 0 \qquad u(4,y) = \sin\left(\frac{\pi y}{4}\right) + 5\sin\left(\frac{\pi y}{12}\right)$$

Solution:

If we find the first and second derivatives of the product solution,

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2 v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2 w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2 v}{dx^2}w + v\frac{d^2 w}{dy^2} = 0$$

then we can separate variables.

$$\frac{d^2 v}{dx^2}w = -v\frac{d^2 w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2 v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2 w}{dy^2}$$

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Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \lambda$$

$$\frac{d^2v}{dx^2} = \lambda v(x)$$

$$\frac{d^2v}{dx^2} - \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

$$\frac{d^2w}{dy^2} = -\lambda w(y)$$

$$\frac{d^2w}{dy^2} + \lambda w = 0$$

Our boundary conditions are $w(0) = 0$, $w'(6) = 0$, and $v'(0) = 0$, and the solution to the second ordinary differential equation boundary value problem is

$$w(y) = c_1 \cos(\sqrt{\lambda}y) + c_2 \sin(\sqrt{\lambda}y)$$

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$$w(0) = 0, \text{ so } c_1 = 0$$

$$w'(6) = -c_1\sqrt{\lambda}\sin(6\sqrt{\lambda}) + c_2\sqrt{\lambda}\cos(6\sqrt{\lambda}) = 0$$

and the solution is

$$\cos 6\sqrt{\lambda} = 0$$

$$\lambda_n = \left(\frac{\frac{\pi}{2} + n\pi}{6}\right)^2 = \left(\frac{\pi + 2n\pi}{12}\right)^2$$

$$w(y) = c_2 \sin\frac{(\pi + 2n\pi)y}{12} \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the first equation, we get

$$v(x) = c_3 e^{-\sqrt{\lambda}x} + c_4 e^{\sqrt{\lambda}x}$$

Equivalently, this can be rewritten as

$$v(x) = c_3 \sinh(\sqrt{\lambda}x) + c_4 \cosh(\sqrt{\lambda}x)$$

which allows us to apply $v'(0) = 0$. When we do, we find $c_3 = 0$, and the solution is

$$v(x) = c_4 \cosh(\sqrt{\lambda}x)$$

$$v(x) = c_4 \cosh\left(\frac{(\pi + 2n\pi)x}{12}\right)$$

Then the product solution is

$$u(x, y) = v(x)w(y)$$

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$$u(x,y) = c_4 \cosh\left(\frac{(\pi + 2n\pi)x}{12}\right) c_2 \sin\left(\frac{(\pi + 2n\pi)y}{12}\right)$$

$$u_n(x,y) = B_n \cosh\left(\frac{(\pi + 2n\pi)x}{12}\right) \sin\left(\frac{(\pi + 2n\pi)y}{12}\right) \qquad n = 1, 2, 3, \ldots$$

$$u(x,y) = \sum_{n=1}^{\infty} B_n \cosh\left(\frac{(\pi + 2n\pi)x}{12}\right) \sin\left(\frac{(\pi + 2n\pi)y}{12}\right)$$

Substitute

$$u(4,y) = \sin\left(\frac{\pi y}{4}\right) + 5\sin\left(\frac{\pi y}{12}\right)$$

to get

$$\sum_{n=1}^{\infty} B_n \cosh\left(\frac{(\pi + 2n\pi)}{3}\right) \sin\left(\frac{(\pi + 2n\pi)y}{12}\right) = \sin\left(\frac{\pi y}{4}\right) + 5\sin\left(\frac{\pi y}{12}\right)$$

$$B_0 = \frac{5}{\cosh\left(\frac{\pi}{3}\right)}, B_1 = \frac{1}{\cosh \pi}$$

Other than B_0 and B_1, we know $B_2 = B_3 = \ldots = B_n = 0$. Then the solution is

$$u(x,y) = \frac{5}{\cosh\left(\frac{\pi}{3}\right)} \cosh\left(\frac{\pi x}{12}\right) \sin\left(\frac{\pi y}{12}\right) + \frac{1}{\cosh \pi} \cosh\left(\frac{\pi x}{4}\right) \sin\left(\frac{\pi y}{4}\right)$$

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■ 5. Use the product solution to find $u_1(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq 1$ and $0 \leq y \leq 1$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = x \qquad u(0,y) = 0$$

$$u(x,1) = 0 \qquad u(1,y) = \sin(3\pi y)$$

Solution:

We can say that $u = u_1 + u_2$, where u_1 is the solution of

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = x \qquad u(0,y) = 0$$

$$u(x,1) = 0 \qquad u(1,y) = 0$$

and u_2 is the solution of

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u(0,y) = 0$$

$$u(x,1) = 0 \qquad u(1,y) = \sin(3\pi y)$$

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Let's start with u_1. If we find the first and second derivatives of the product solution,

$$u_1(x,y) = v(x)w(y)$$

$$\frac{\partial u_1}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u_1}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u_1}{\partial x^2} = \frac{d^2v}{dx^2}w \qquad \frac{\partial^2 u_1}{\partial y^2} = v\frac{d^2w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

then we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

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$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

Our boundary conditions are $w(1) = 0$, $v(0) = 0$, and $v(1) = 0$, and the solution to the first ordinary differential equation boundary value problem is

$$\lambda_n = (n\pi)^2 \qquad v_n(x) = C\sin(n\pi x) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the second equation, we get

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

$$w(1) = 0$$

$$w(y) = c_1 e^{-\sqrt{\lambda}y} + c_2 e^{\sqrt{\lambda}y}$$

Equivalently, this can be rewritten as

$$w(y) = c_1 \sinh\sqrt{\lambda}(y-1) + c_2 \cosh\sqrt{\lambda}(y-1)$$

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which allows us to apply $w(1) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$w(y) = c_1 \sinh \sqrt{\lambda}(y - 1)$$

$$w(y) = c_1 \sinh n\pi(y - 1)$$

Then the product solution is

$$u_1(x, y) = v(x)w(y)$$

$$u_1(x, y) = C \sin(n\pi x)c_1 \sinh n\pi(y - 1)$$

$$u_1(x, y) = \sum_{n=1}^{\infty} B_n \sin(n\pi x)\sinh n\pi(y - 1)$$

Substitute $u(x,0) = x$.

$$\sum_{n=1}^{\infty} B_n \sin(n\pi x)\sinh(-n\pi) = x$$

Find B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = 2\int_0^1 x \sin(n\pi x)\,dx$$

Integrate, using integration by parts with $u = x$, $du = dx$, $dv = \sin n\pi x\ dx$, and $v = -(1/n\pi)\cos(n\pi x)$.

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$$B_n = 2\left(-\frac{1}{n\pi} x\cos(n\pi x)\Big|_0^1 + \frac{1}{n\pi}\int_0^1 \cos(n\pi x)\, dx \right)$$

$$B_n = 2\left(-\frac{1}{n\pi}(-1)^n + \frac{1}{(n\pi)^2}\sin(n\pi x)\Big|_0^1 \right) = 2\left(-\frac{1}{n\pi}(-1)^n + 0 \right)$$

$$B_n = \frac{2(-1)^{n+1}}{n\pi}$$

Then the solution is

$$u_1(x,y) = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n\pi\sinh(n\pi)} \sin(n\pi x)\sinh n\pi(y-1)$$

Solve for u_2,

$$\frac{\partial u_2}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u_2}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u_2}{\partial x^2} = \frac{d^2v}{dx^2}w \qquad \frac{\partial^2 u_2}{\partial y^2} = v\frac{d^2w}{dy^2}$$

then plug the second derivatives into Laplace's equation,

$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

so that we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

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$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \lambda$$

$$\frac{d^2v}{dx^2} = \lambda v(x)$$

$$\frac{d^2v}{dx^2} - \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

$$\frac{d^2w}{dy^2} = -\lambda w(y)$$

$$\frac{d^2w}{dy^2} + \lambda w = 0$$

Our boundary conditions are $w(1) = 0$, $v(0) = 0$, and $v(1) = 0$, and the solution to the second ordinary differential equation boundary value problem is

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$$\lambda_n = (n\pi)^2 \qquad w_n(y) = C\sin(n\pi y) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the first equation, we get

$$\frac{d^2v}{dx^2} - \lambda v = 0$$

$$v(x) = c_1 e^{-\sqrt{\lambda}x} + c_2 e^{\sqrt{\lambda}x}$$

Equivalently, this can be rewritten as

$$v(x) = c_1 \sinh(\sqrt{\lambda}x) + c_2 \cosh(\sqrt{\lambda}x)$$

which allows us to apply $v(0) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$v(x) = c_1 \sinh(\sqrt{\lambda}x)$$

$$v(x) = c_1 \sinh(n\pi x)$$

Then the product solution is

$$u_2(x, y) = v(x)w(y)$$

$$u_2(x, y) = c_1 \sinh(n\pi x) C \sin(n\pi y)$$

$$u_2(x, y) = \sum_{n=1}^{\infty} B_n \sinh(n\pi x)\sin(n\pi y)$$

Substitute $u(1,y) = \sin(3\pi y)$.

$$\sum_{n=1}^{\infty} B_n \sinh(n\pi)\sin(n\pi y) = \sin(3\pi y)$$

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$$B_3 = \frac{1}{\sinh(3\pi)}$$

Other than B_3, we know $B_1 = B_2 = B_4 = \ldots = B_n = 0$. Then the solution is

$$u_2(x,y) = \frac{1}{\sinh(3\pi)} \sinh(3\pi x)\sin(3\pi y)$$

And therefore,

$$u(x,y) = \sum_{n=1}^{\infty} \frac{2(-1)^n}{n\pi} \sin(n\pi x)\sinh \pi n(y-1) + \frac{1}{\sin(3\pi)} \sinh(3\pi x)\sin(3\pi y)$$

■ 6. Use the product solution to find $u_1(x,y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq 2\pi$ and $0 \leq y \leq 2\pi$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = \cos x \qquad u_x(0,y) = 0$$

$$u(x,2\pi) = \cos x \qquad u_x(2\pi, y) = 0$$

Solution:

We can say that $u = u_1 + u_2$, where $u_1$is the solution of

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$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = \cos x \qquad u_x(0,y) = 0$$

$$u(x,2\pi) = 0 \qquad u_x(2\pi, y) = 0$$

and u_2 is the solution of

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u_x(0,y) = 0$$

$$u(x,2\pi) = \cos x \qquad u_x(2\pi, y) = 0$$

Lets start with u_1. If we find the first and second derivatives of the product solution,

$$u_1(x,y) = v(x)w(y)$$

$$\frac{\partial u_1}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u_1}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u_1}{\partial x^2} = \frac{d^2 v}{dx^2}w \qquad \frac{\partial^2 u_1}{\partial y^2} = v\frac{d^2 w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2 v}{dx^2}w + v\frac{d^2 w}{dy^2} = 0$$

then we can separate variables.

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$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

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The boundary conditions are $w(2\pi) = 0$, $v'(0) = 0$, and $v'(2\pi) = 0$, and the solution to the first ordinary differential equation boundary value problem is

$$\lambda_0 = 0 \qquad\qquad v_0 = C$$

$$\lambda_n = \left(\frac{n\pi}{2\pi}\right)^2 = \frac{n^2}{4} \qquad v_n(x) = C\cos\left(\frac{nx}{2}\right) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the second equation, we get

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

For $\lambda_0 = 0$, the solution is

$$w_0(y) = c_1 + c_2(y - 2\pi)$$

$w(2\pi) = 0$, so $c_1 = 0$ and $w_0(y) = c_2(y - 2\pi)$

For $\lambda_n = n^2/4$, the solution is

$$w(y) = c_1 e^{-\sqrt{\lambda}y} + c_2 e^{\sqrt{\lambda}y}$$

Equivalently, this can be rewritten as

$$w(y) = c_1 \sinh\sqrt{\lambda}(y - 2\pi) + c_2 \cosh\sqrt{\lambda}(y - 2\pi)$$

which allows us to apply $w(2\pi) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$w(y) = c_1 \sinh\sqrt{\lambda}(y - 2\pi)$$

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$$w(y) = c_1 \sinh\left(\frac{n(y-2\pi)}{2}\right)$$

Then the product solution is

$$u_1(x,y) = v(x)w(y)$$

$$u_1(x,y) = A_0(y-2\pi) + \sum_{n=1}^{\infty} A_n \cos\left(\frac{nx}{2}\right) \sinh\left(\frac{n(y-2\pi)}{2}\right)$$

Substitute $u_1(x,0) = \cos x$.

$$A_0(-2\pi) + \sum_{n=1}^{\infty} A_n \cos\left(\frac{nx}{2}\right) \sinh(-n\pi) = \cos x$$

$$A_2 = -\frac{1}{\sinh(2\pi)}$$

Other than A_2, we know $A_0 = A_1 = A_3 = \ldots = A_n = 0$. Then the solution is

$$u_1(x,y) = -\frac{1}{\sinh(2\pi)} \cos x \sinh(y-2\pi)$$

Solve for u_2,

$$\frac{\partial u_2}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u_2}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u_2}{\partial x^2} = \frac{d^2v}{dx^2}w \qquad \frac{\partial^2 u_2}{\partial y^2} = v\frac{d^2w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

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$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

so that we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

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$$\frac{d^2w}{dy^2} - \lambda w = 0$$

The boundary conditions are $w(2\pi) = 0$, $v'(0) = 0$, and $v'(2\pi) = 0$, and the solution to the first ordinary differential equation boundary value problem is

$$\lambda_0 = 0 \qquad\qquad v_0 = C$$

$$\lambda_n = \left(\frac{n\pi}{2\pi}\right)^2 = \frac{n^2}{4} \qquad v_n(x) = C\cos\left(\frac{nx}{2}\right) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the second equation, we get

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

For $\lambda_0 = 0$, the solution is

$$w_0(y) = c_1 + c_2 y$$

$w(0) = 0$, so $c_1 = 0$ and $w_0(y) = c_2 y$

For $\lambda_n = n^2/4$, the solution is

$$w(y) = c_1 e^{-\sqrt{\lambda}y} + c_2 e^{\sqrt{\lambda}y}$$

Equivalently, this can be rewritten as

$$w(y) = c_1 \sinh(\sqrt{\lambda}y) + c_2 \cosh(\sqrt{\lambda}y)$$

which allows us to apply $w(0) = 0$. When we do, we find $c_2 = 0$, and the solution is

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$$w(y) = c_1 \sinh(\sqrt{\lambda} y)$$

$$w(y) = c_1 \sinh\left(\frac{ny}{2}\right)$$

Then the product solution is

$$u_2(x, y) = v(x)w(y)$$

$$u_2(x, y) = A_0 y + \sum_{n=1}^{\infty} A_n \cos\left(\frac{nx}{2}\right) \sinh\left(\frac{ny}{2}\right)$$

Substitute $u_2(x,2\pi) = \cos x$.

$$A_0(2\pi) + \sum_{n=1}^{\infty} A_n \cos\left(\frac{nx}{2}\right) \sinh(n\pi) = \cos x$$

$$A_2 = \frac{1}{\sinh(2\pi)}$$

Other than A_2, we know $A_0 = A_1 = A_3 = \ldots = A_n = 0$. Then the solution is

$$u_2(x, y) = \frac{1}{\sinh(2\pi)} \cos x \sinh y$$

And therefore,

$$u(x, y) = -\frac{1}{\sinh(2\pi)} \cos x \sinh(y - 2\pi) + \frac{1}{\sinh(2\pi)} \cos x \sinh y$$

$$u(x, y) = \frac{\cos x}{\sinh(2\pi)}(\sinh y - \sinh(y - 2\pi))$$

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